

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 15

Algorithmic Complexity and the Search for Generative Equations

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

1. Kolmogorov complexity is

define it in one sentence

2. A string is “random” (Kolmogorov) if

Compare to Shannon randomness (maximum entropy):

3. Shannon entropy vs. Kolmogorov complexity

Shannon: property of a _____

Kolmogorov: property of an _____

Shannon requires knowledge of:

Kolmogorov is:

4. The machine underneath

The Church-Turing thesis says:

A universal machine is:

The halting problem is unsolvable because (sketch the saboteur):

5. Why Kolmogorov complexity is uncomputable

sketch the Berry-paradox argument in your own words

The diagonal maneuver has now appeared four times:

Cantor (Ch 2): _____ Gödel (Ch 4): _____

Halting (this ch): _____ K (this ch): _____

6. A generative equation is a _____ that produces _____.

Finding it is an act of _____.

7. The search space problem

Fixed grammar:

Variable grammar (the frontier):

The expressiveness-searchability trade-off:

8. The hierarchy of ambition

Statistics $\rightarrow$ ML $\rightarrow$ Symbolic regression $\rightarrow$ Solomonoff induction

Each level adds:

Each level costs:

9. Where my original intuition about compression sits on this map

locate it precisely
